

BRIAN (YUAN-AN) LIU

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GEN AI & CYBERSECURITY SPECIALIST

Strategic technologist with 10+ years of experience designing secure, scalable cloud architectures and enterprise-grade Generative AI solutions. Leveraging a deep background in Staff Software Engineering (Zscaler, AWS, F5) to bridge the gap between complex business requirements and technical implementation. Expert in orchestrating LLM workflows (RAG, Agents), optimizing cloud infrastructure for cost and latency, and implementing security principles. Passionate about shifting from manual development cycles to automated (agentic workflows), high-level architectural design that drives business innovation and operational efficiency.

EDUCATION

ILLINOIS INSTITUTE OF TECHNOLOGY, Chicago, IL

Master of Science, Computer Science

Aug 2016 - May 2018

Association: IIT-Security Lab

CHUNG YUAN CHRISTIAN UNIVERSITY, Taoyuan, Taiwan

Bachelor of Science, Computer Science

Sep 2011 - Jun 2015

Association: CYCU-Software Engineering Lab

SKILLS

- **Languages:** Go, Python, C++, C, Shell Script, TypeScript, Java
- **Cloud & Infrastructure:** AWS, Azure, GCP, Multi-Cloud, CDK, Terraform (IaC), Kubernetes, Docker, Helm., Nginx, Bedrock, SDN, OVS, eBPF, GNU toolchain
- **AI/ML/Security:** Gen AI, LLM, ML, RAG, NLP, RAGAS, LangChain, Hugging Face, GPT, Claude, Gemini, LLaMA, NLP, Distributed/Embedded Systems, Cybersecurity, BMaaS, TensorFlow, PyTorch
- **Networking/Protocols:** 5G, gRPC, NSH, SFC, VLAN, VxLAN, TCP/IP, REST, SOAP, P2P, ZMQ, SRv6
- **Data & Services:** DynamoDB, PostgreSQL, MySQL, OpenSearch, Git, Kafka, Kinesis, Protobuf, vim, Telemetry, Prometheus, Temporal

EXPERIENCE

ZSCALER, INC. San Jose (Remote), CA

Nov 2024 - present

Senior Software Development Engineer

- Leading technical solution architecture and implementation for a new cloud-native security CTEM service, utilizing the Zscaler data fabric for high-availability and strategic customer integration.
- Optimizing cloud-native infrastructure deployment costs through detailed cost analysis and capacity planning, resulting in projected revenue models for emerging AI/ML service offerings.
- Engineering and deploying production-grade AI workflows for real-time threat log analysis using RAG-based agents, incorporating diverse LLMs (cloud, on-prem, self-hosted) and continuous model training pipelines.
- Designing and scaling core Zscaler Internet Access (ZIA) inventory services utilizing container orchestration (Kubernetes and Docker) to ensure high throughput and scalability for global cloud services.
- Spearheading advanced R&D initiatives across AI and Cybersecurity to define the technical roadmap and future cloud-native security offerings, ensuring architectural alignment with market needs.
- Inventor and author on multiple Generative AI, ML, and cybersecurity patents, establishing technical thought leadership within the organization and the broader security industry.
- Driving internal engineering adoption and defined best practices for integrating cutting-edge AI/ML technologies, including active contribution to relevant open-source AI projects.

AMAZON WEB SERVICES, INC. (AWS), Sunnyvale, CA

Nov 2019 - Nov 2024

Software Development Engineer 2

Software Development Engineer 1

- Architected and developed AI-powered cloud security services using Generative AI, establishing technical thought leadership across ML and cloud security patents.
- Provided technical leadership for engineering teams, driving business planning, defining product requirements with key stakeholders, and delivering high-impact solutions for large-scale product launches.
- Optimized critical infrastructure cost-efficiency through analysis and collaborated with Product Management on revenue projections for new AWS network service offerings.

- Led end-to-end development of critical AWS network services (Private 5G, VPC, Site2Site VPN, Telco Network Builder) and managed hardware acceleration integration (Outposts, EC2, GPUs) in physical data centers.
- Engineered high-throughput data streaming solutions for AWS Aurora, developing an advanced BinLog manipulation library to ensure low-latency replication and consistency.
- Enhanced service reliability and operational excellence by establishing automation pipelines and leading Root Cause Analysis (RCA) efforts for critical AWS services.

F5 NETWORKS, INC, San Jose, CA

Jul 2018 – Nov 2019

Software Engineer 2

Software Engineer 1

- Built policy enforcement manager, service function chaining, and modular solutions for F5 BIG-IP.
- Integrated application delivery networking into CI/CD, conducted SDN controller and RFC feasibility studies.
- Collaborated cross-functionally to deliver innovative products, provided technical leadership and mentoring.

KYPHER, LLC, St. Louis, MO

May 2017 - Aug 2017

Application Developer Intern

- Full Stack Developer for Kypher™ HIPAA Compliant Messaging application.

R.O.C. NAVY, FFG-934, Taiwan

Jul 2015 - Jun 2016

Sonar Technician Seaman (Mandatory Military Service)

- Operated and maintained sonar and underwater fire control systems.

MOREMOTE, INC., Taipei, Taiwan

Jul 2014 - Jun 2015

Software Engineer

- Developed applications for two IP-Camera projects (BBCam™ and Ability HomeCam™).
- Implemented microservices with containerization, REST, Kafka with P2P technology.
- Performed strategic assessments, feasibility studies, software release maintenance.
- Worked directly with OEM clients to provide technical consultation and solution delivery.

PUBLICATION

- **A Distributed Virtual Time on Embedded System for Evaluating Cyber-Physical Systems** (ACM SIGSIM-PADS 2019)
(*Best Paper Award*)
- **A Distributed Embedded Linux System Smart Grid Testbed** (GCASR 2018)

PATENT

- **Self-healing and resiliency in radio-based networks using a community model.** Dec 2022
US 11,888,701 B1, Issued Jan 2024
- **Enhancing availability of radio-based applications using multiple compute instances and virtualized network function accelerators at cloud edge** Aug 2022
US 11,937,103 B1, Issued Mar 2024
- **Hybrid Automotive Cybersecurity Monitoring Service** Feb 2025
US 2026/0247154 A1, Pending Aug 2026
- **Confidential (AI)** Jun 2026
Patent pending
- **Confidential (AI)** Jun 2026
Patent pending
- **Confidential (AI)** May 2025
Patent pending
- **Confidential** Dec 2022
Patent pending